CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Cardboard joins and careful cutting

Goal: I can make a stable cardboard connection and test it before adding decoration.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 4

Ask | Plan | Build | Test | Improve

You are done when

Show one stable connection and explain why it holds.

You will use

- Clean cardboard
- Paper
- Pencils
- Rulers
- Student scissors
- Masking tape
- Reusable trays
- Teacher-prepared pieces if needed
- Optional approved ChompSaw station

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Read today's goal.
- Look at the example.

2 Step 2: Tell your partner — 1 min

- Tell your partner one useful detail.

3 Step 3: Watch and try — 5 min

- Watch the teacher cut away from the holding hand.
- Compare a tab, slot, and taped join.
- Watch the teacher test each join gently.

4 Step 4: Make it work — 17 min

- Sketch a small shape that stands by itself.
- Choose a tab, slot, or taped join.
- Cut with approved scissors or use teacher-prepared pieces. Assemble your shape.

5 Step 5: Check one thing — 5 min

- Place your shape on the table and test its balance.
- Change one weak join.
- Compare the first and improved versions.

6 Step 6: Reset together — 3 min

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

7 Step 7: Show your result — 2 min

- Show one stable connection and explain why it holds.

What to turn in

Three labeled joint samples and one stable shape in the class tray.

No computer? Another way to do it

Paper and cardstock tabs and slots work for every step.

Finished early? Next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Stay safe

Scissors are the default. A ChompSaw station opens only after school approval and the exact machine demonstration.

Grade targets

Grade 5: Build one stable shape using a demonstrated join.
Grade 6: Compare two join types and explain the tradeoff before choosing.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.